Chapter 3
Understanding the Supply Process: Evaluating Process Capacity

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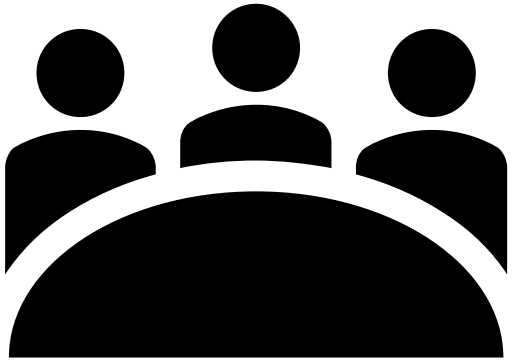
Collaborative Clarification

- Students raise questions about parts they found difficult
- Any students may answer and one additional student may add comments
- The professor clarifies both the questions and answers if necessary
- Students who raised questions and answered write down what they did (A Q&A sheet)



The Number of Stations = 1

- Assumptions: 1 operator assembles 1 vehicle = 4hr (ST: Standard Time)
- CT = 4hr
  - If you make only 2 units a day, it is OK
  - But what if the demand is more than two?

Assemble Time = 4hr

CT = 4hr

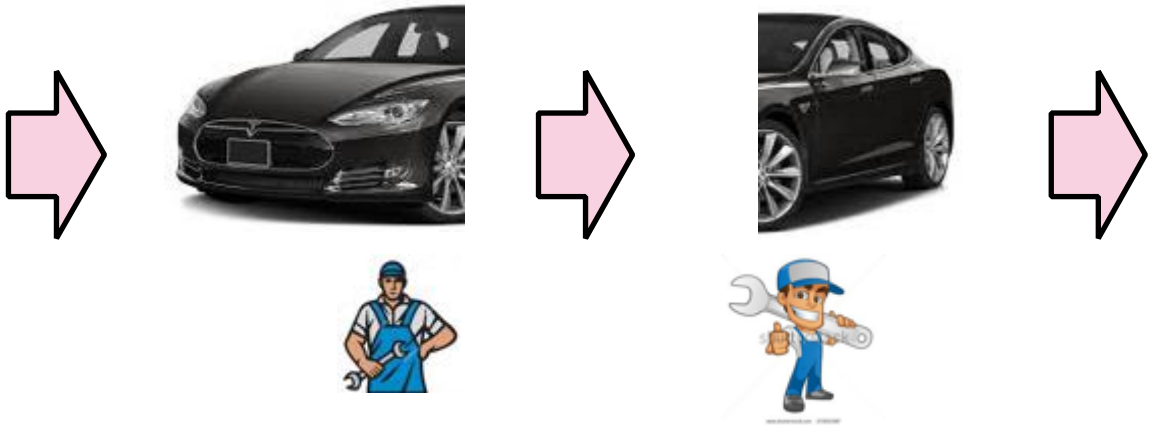
How to Meet Demand?

- Takt time (TT)
  - A company would build an assembly line that produces 60 cars a day (8h=480m)
  - Takt time $= \frac{Available\ Time}{Demand}$ =480/60=8m (A car every 8m)
- So, ideal station's cycle time = 8min, and ST = 240min
  - The number of stations = 240/8 = 30
- Let's try to decompose a car assembly task into smaller tasks to understand an assembly design

The Number of Stations = 2

Assemble Time = 4

Assemble Time = 2 Assemble Time = 2



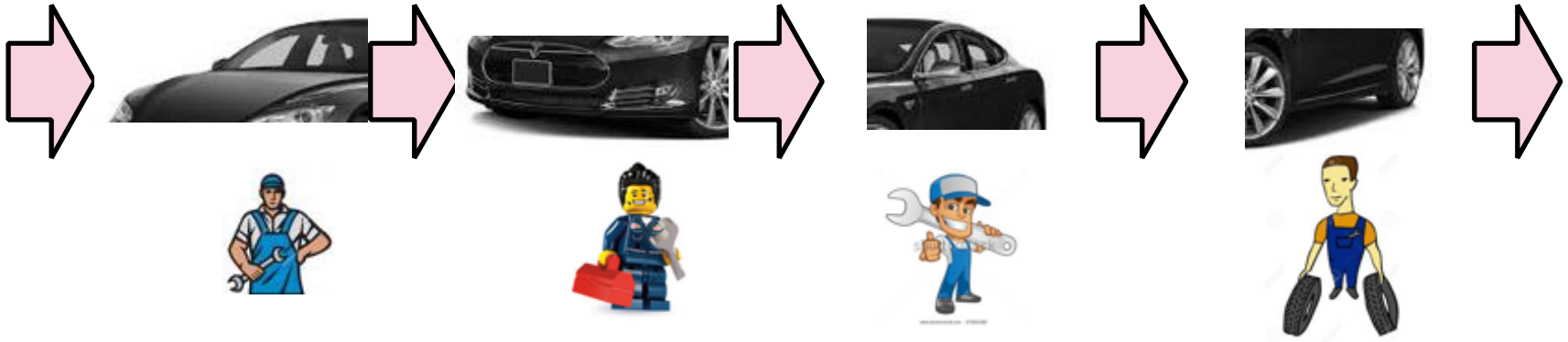
How many cars are produced a day?

- ❑ Each station's CT is 2hr and then Then the line's CT is 2hr
- ❑ Therefore, the number of cars assembled a day = 8÷2=4

The Number of Stations = 4

Assemble Time = 4

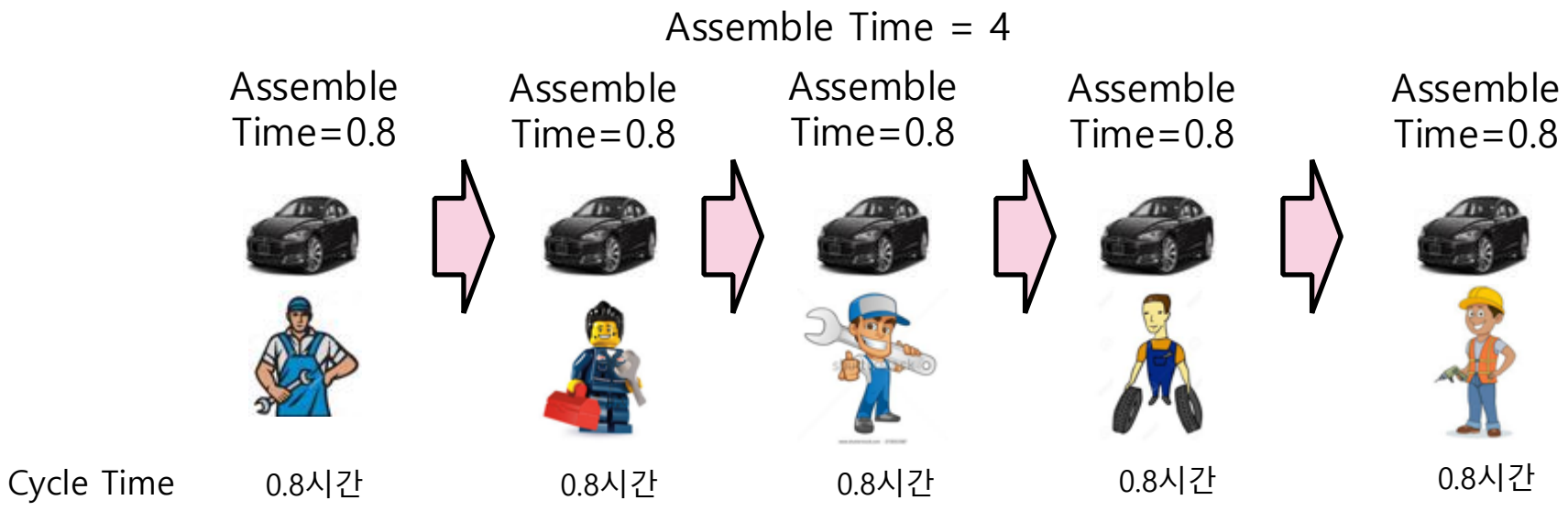
Assemble Time=1 Assemble Time=1 Assemble Time=1 Assemble Time=1



How many cars are produced a day?

- ❑ Each station's CT is 1hr and then the line's CT is 1hr
- ❑ Therefore, the number of cars assembled a day = 8÷1=8

The Number of Stations = 5



How many cars are produced per day?

- ❑ Assume each station's CT is 0.8hr, and then the line's CT is 0.8hr
- ❑ Therefore, the number of cars assembled a day = 8÷0.8=10
- ❑ **Although the task can be decomposed continuously, it will be difficult to make each station take the same working time**

The Number of Stations = 5 (Different CTs)

Assemble Time = 4

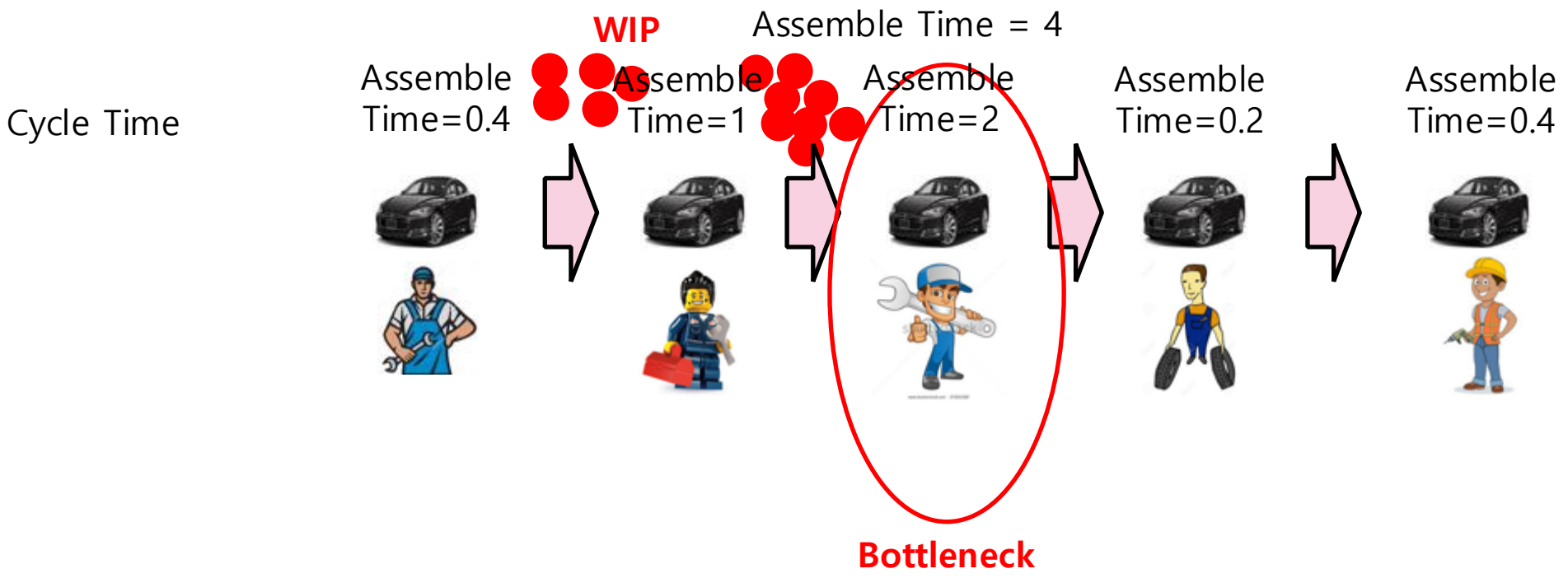
Cycle Time

Assemble Time=0.4 → Assemble Time=1 → Assemble Time=2 → Assemble Time=0.2 → Assemble Time=0.4

How many cars are produced per day?

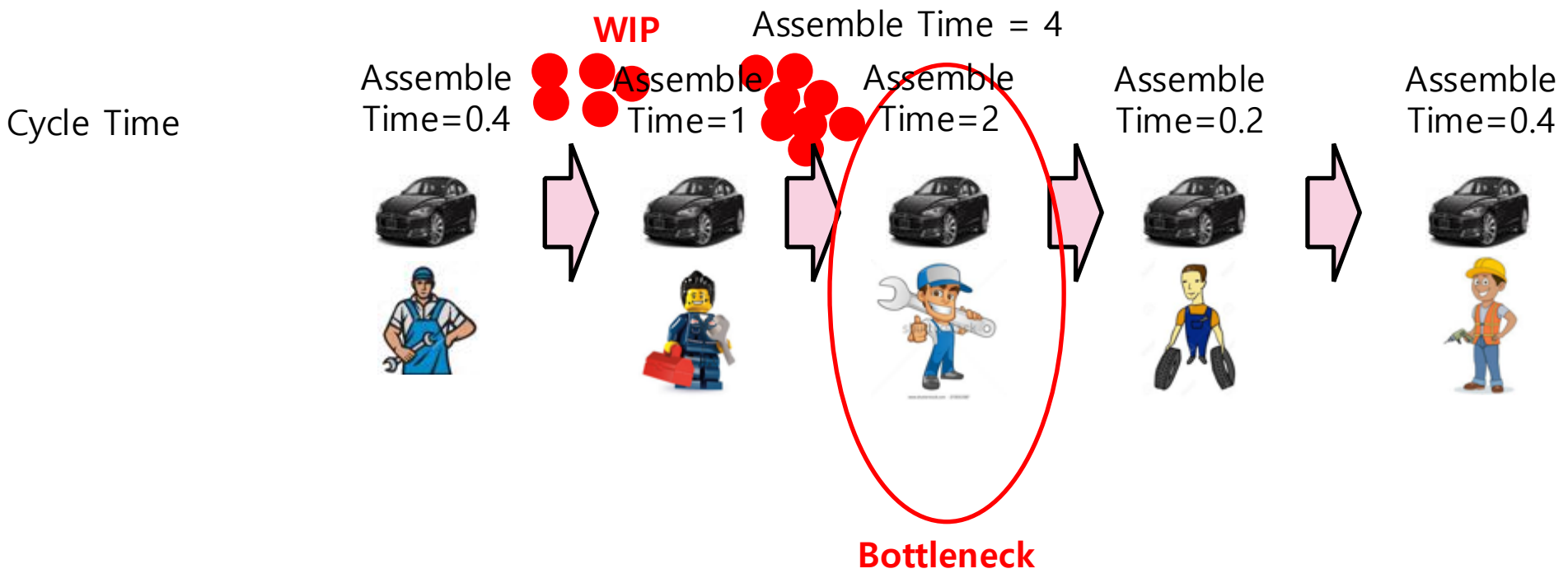
- ❑ The line's CT is 2hr (Bottle neck)
- ❑ Therefore, the number of cars assembled a day = 8÷2=4

The Number of Stations = 5 (Different CTs)



WIP will be increasing in front of the bottleneck station

The Number of Stations = 5 (Different CTs)



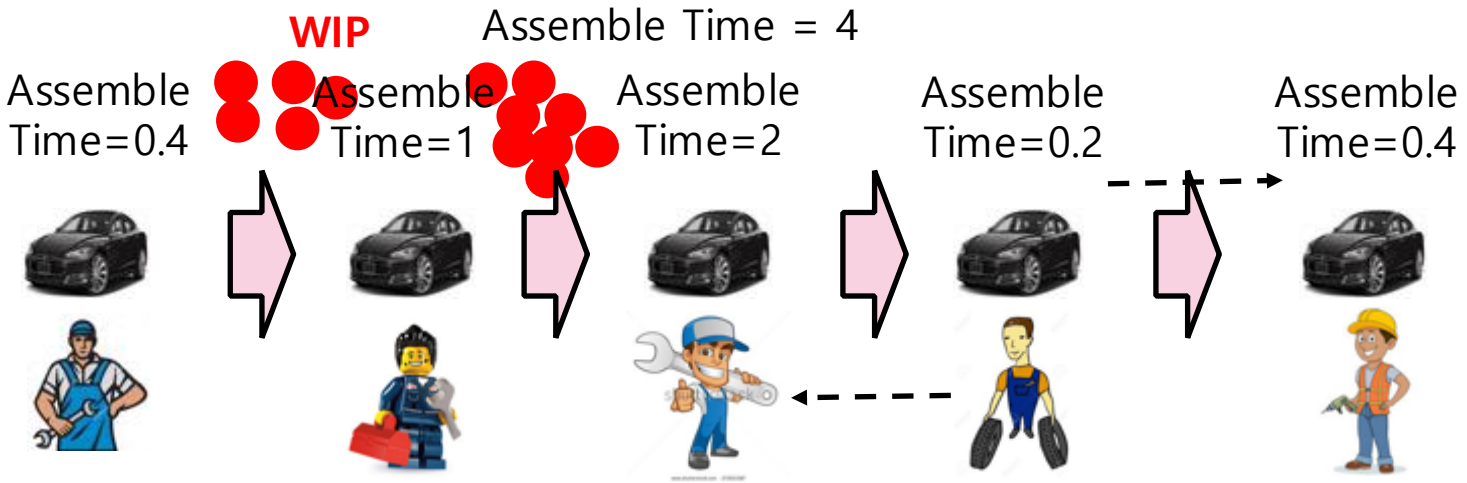
Line's CT = 2hr, therefore, daily production amount is 4

Average Flow Time = Average WIP ↑ x Average Line CT = (2hr±?)

Due to the low LOB, WIP increases, causing the average FT to increase

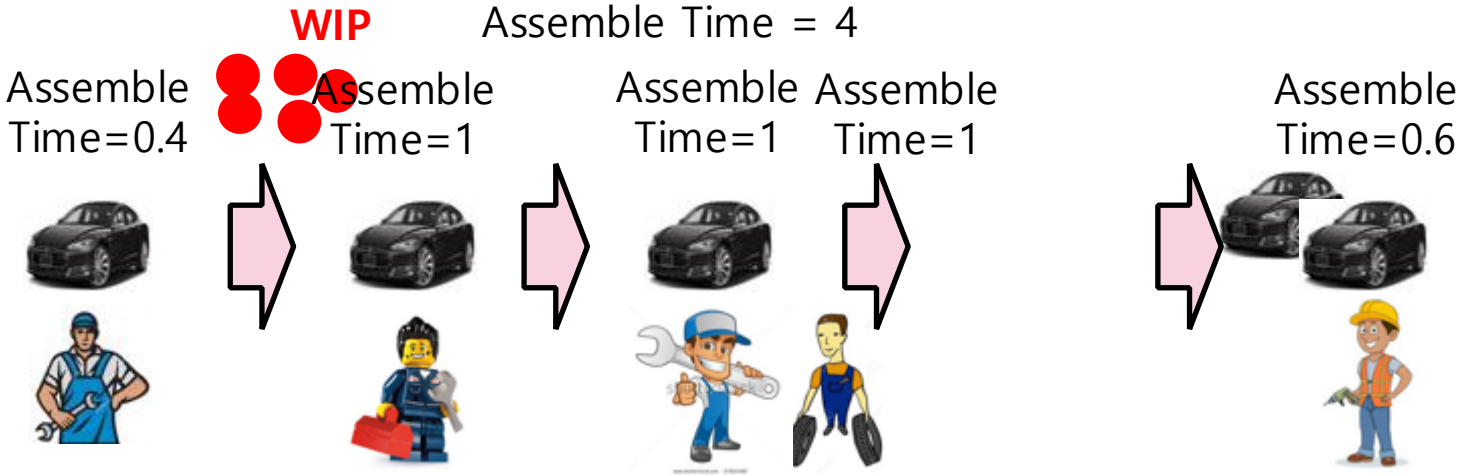
Line Balancing

Cycle Time



$$\text{LoB} = \frac{0.4+1+1+1+0.6}{1\ X\ 5} = \frac{4}{5} = 80\%$$

Cycle Time

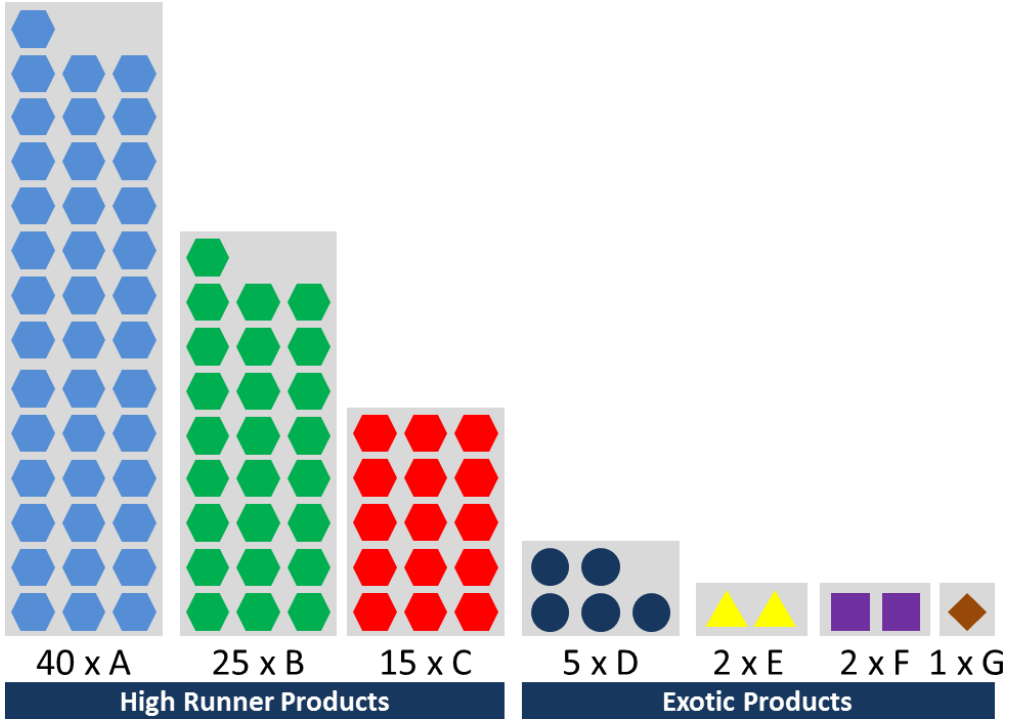


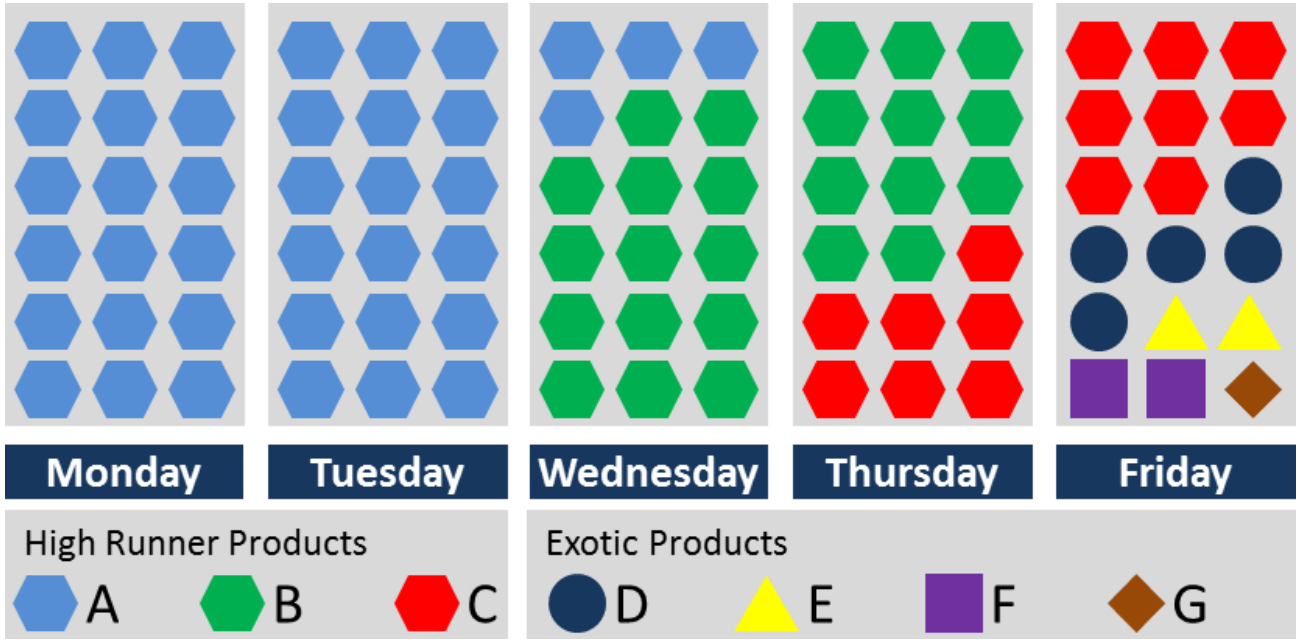
How to Manage Mass Production

- Shorten assembly time with learning effect through repetitive work
  - Introduction of large batch size and part standardization policy
  - It lacks responsiveness to customers with small quantities or customized products
  - In addition, incorrect process settings can cause all products in the batch to be defective
- For maximum productivity, inspection is done at the end of the line
  - Calculate the line defect rate (because of the loss) and determine the additional material input
  - It is difficult to detect defects early, so it is difficult to solve the root causes
- Adopt a push policy for maximum output
  - If a worker becomes less efficient, Line Balancing collapses
  - The Bottle Neck process is stocked, and the subsequent process is starved

Revisit: Mass Production – Excess Inventory

- **Prompt**: This is a typical mass production schedule. Describe the implications of this schedule.



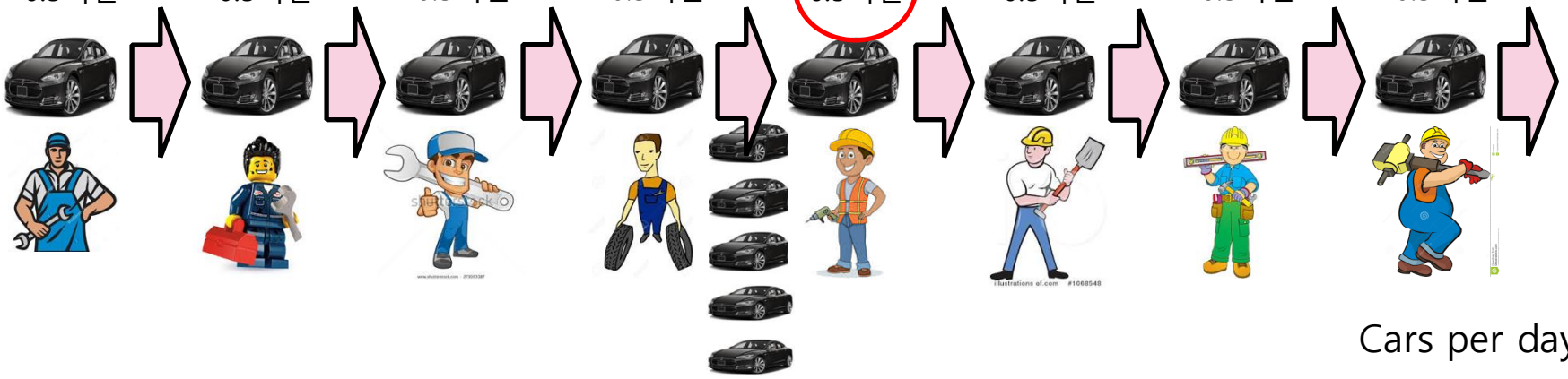


Side Effect 1 – Bottle Neck

Assemble Time = 4

0.6시간

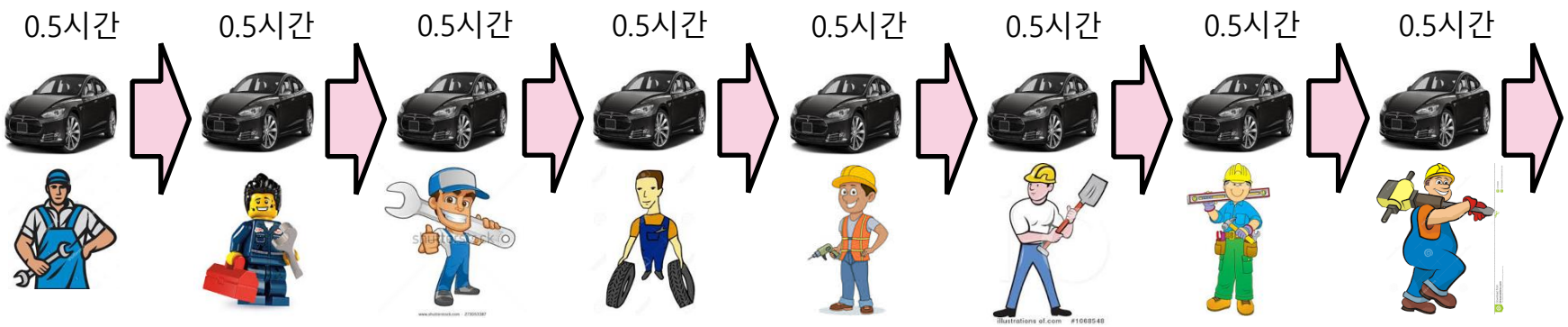
0.5시간 0.5시간 0.5시간 0.5시간 0.5시간 0.5시간 0.5시간 0.5시간



Cars per day = 16대

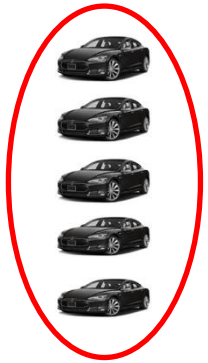
Side Effect 2 – Over-Production

Assemble Time = 4



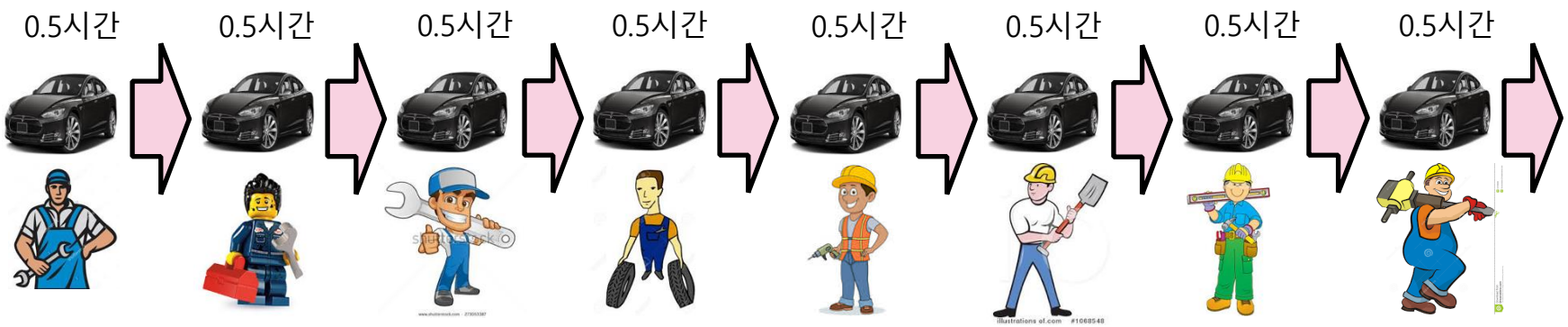
Cars per day = 16대

10% defective from historical data, so 110% production

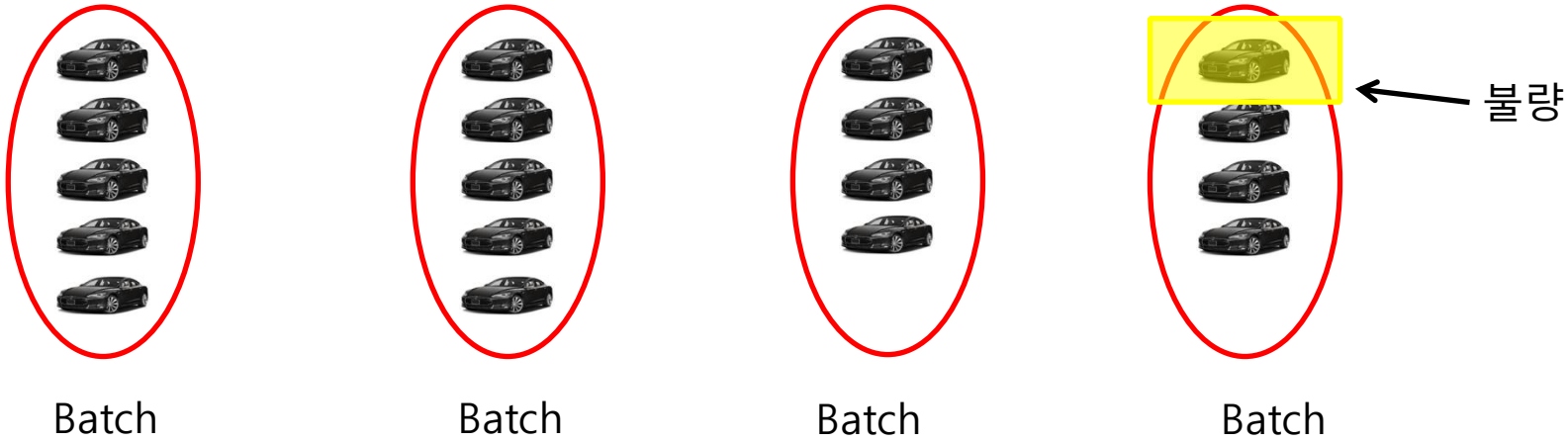


Side Effect 3 – A Single Defective causes Whole Defective

Assemble Time = 4



Cars per day = 16대



Formative Quiz

- Begin with a short formative quiz based on the audio overview to assess comprehension
- Students complete a quiz sheet and submit it (A quiz sheet)

Applied Learning

- Distribute a short case and its accompanying podcast if necessary
  - POM3 (Applied Learning) How Capacity Utilization Can & Should Drive Pricing Decisions
- Students would share their thoughts with peers
- Students should write down their reflections (A reflection sheet)

Debating Statement

- Should Operations Managers be cautious when using simplified flow diagrams, since they may hide the real bottlenecks and capacity constraints of complex processes?
  - Are simple flowcharts dangerous? Do they make a process look smooth on paper while hiding the actual traffic jams and limits that happen in real life?
  - Managers should not trust simple diagrams blindly. Just as a highway map doesn't show you the potholes or accidents, a simple process map hides the messy reality of where work actually gets stuck.
  - Be careful with clean diagrams. They usually show how work should flow (the 'Happy Path'), but they fail to show the rework, errors, and delays that determine your actual capacity.

- **The "YES" Argument (Focus: Hidden Complexity & Variance)**
  - *The map is not the territory; the details kill you.*
- **The "NO" Argument (Focus: Clarity & Strategy)**
  - *You need a compass, not a satellite photo.*

Debating Statement

- In Operations Management, a supply-constrained environment should be designed around the bottleneck. If other resources show large utilization gaps compared to the bottleneck, it typically reflects a system design problem rather than a natural outcome of process balancing.
  - When you have more customers than products, your factory is limited by its slowest machine (the bottleneck). If that machine is working 100% of the time but every other machine is sitting idle for half the day, you have designed the system poorly.
  - In a busy factory, the bottleneck dictates your speed. If the bottleneck is busy but other resources have huge gaps of free time, it's not a 'balanced' system—it's a waste of money. You bought too much capacity for the non-essential steps.
  - Design your system around the weakest link. If the weakest link is straining to hold on, but the other links are thick enough to hold an elephant, you have over-engineered the rest of the chain. That is a design failure, not a natural result.
- **The "YES" Argument (Focus: Lean & Capital Efficiency)**
  - *Idle machines are expensive monuments to bad planning.*
- **The "NO" Argument (Focus: Theory of Constraints & Resilience)**
  - *The gap is not waste; it is insurance.*

Thank You